AURORA-9 ORBITAL GREENHOUSE — OPERATIONS MANUAL (Rev. 4)

Section 1: Station Overview

The Aurora-9 orbital greenhouse is a fictional demonstration habitat used as sample data for the DocuMind project. The station was commissioned in March 2041 and operates in a circular low-Earth orbit at an altitude of 410 kilometers with an orbital inclination of 51.6 degrees.

Aurora-9 hosts a permanent crew of six and cultivates fourteen crop varieties across three pressurized greenhouse modules, designated Botany Bay A, Botany Bay B, and Botany Bay C. Botany Bay A is the largest module at 96 square meters of growing area; modules B and C each provide 64 square meters.

The station's primary mission is to demonstrate closed-loop food production for long-duration spaceflight. Secondary objectives include pharmaceutical plant research and public education outreach. The program is funded through 2049 under the Meridian Accord research charter.

SECTION 2: ENVIRONMENTAL CONTROL

Section 2: Environmental Control Systems

Greenhouse atmosphere is maintained at a carbon dioxide setpoint of 1,200 parts per million during photoperiod hours. Air temperature is held at 22.5 degrees Celsius during the simulated day and lowered to 18.0 degrees Celsius at night. Relative humidity is regulated at 65 percent in all three modules.

Illumination is provided by tunable LED arrays operating on a 16-hour photoperiod. Peak photosynthetic photon flux density is 850 micromoles per square meter per second at canopy height in Botany Bay A, and 700 in modules B and C.

The water reclamation loop recovers 93 percent of transpired and irrigation water. Nutrient solution pH is maintained between 5.8 and 6.2, with electrical conductivity between 1.6 and 2.2 millisiemens per centimeter depending on crop stage. Environmental sensors report every 30 seconds, and any parameter out of range for more than 5 minutes triggers a Tier-2 alarm to the Systems Engineer on duty.

SECTION 3: CREW ORGANIZATION

Section 3: Crew Roles and Scheduling

The six-person crew comprises one Station Commander, one Horticulture Lead, two Systems Engineers, one Medical Officer, and one Data Scientist. The Station Commander holds final authority over all operational decisions and reports to the ground control facility once per day.

Preventive maintenance on the environmental control system is performed every 45 days and requires both Systems Engineers working a full shift. Crop maintenance follows fixed harvest cycles: leaf lettuce is harvested every 28 days, cherry tomatoes every 72 days, and dwarf wheat every 84 days.

Crew rotations last 180 days. Handover between outgoing and incoming crews takes place over a 5-day overlap period. Each crew member is allocated 2.5 hours of daily exercise and one full rest day per week. Psychological support video conferences with ground staff are scheduled every 14 days.

SECTION 4: SAFETY PROCEDURES

Section 4: Safety and Emergency Procedures

Ammonia is used in the external thermal control loop and is the station's highest-consequence hazard. If an ammonia leak alarm sounds, all crew must don protective masks within 60 seconds and evacuate to the Node 2 safe haven, sealing the hatch behind them. The Node 2 module carries 72 hours of independent life support for six people.

Fire suppression in the greenhouse modules uses carbon dioxide flooding rather than water mist, to protect the electrical growing infrastructure. Fire and depressurization drills are conducted monthly, on the first Tuesday of each month.

The emergency return vehicle remains docked at all times, seats all six crew members, and can undock within 4 minutes of an evacuation order. Emergency rations independent of greenhouse production cover 45 days. A medical emergency that exceeds the Medical Officer's onboard capability triggers a return-to-Earth decision within 12 hours.

SECTION 5: COMMUNICATIONS AND DATA

Section 5: Communications and Data Handling

Telemetry is downlinked four times per day through the Ka-band high-gain antenna at a peak rate of 600 megabits per second. Each orbit includes a communications blackout window of approximately 12 minutes when neither relay satellite is visible.

The primary ground station is located in Kirkenes, Norway, with a backup facility in Hobart, Tasmania. Science imagery from the greenhouse modules is captured hourly by 24 fixed cameras and stored onboard for 90 days before archival downlink.

All experiment data is mirrored to two independent storage arrays before deletion is permitted. Flight software updates are uplinked no more than once per 30 days and require sign-off from both the Station Commander and the ground software integrity board. This manual is reviewed annually; Revision 4 was approved in January 2046.